

STUDENT ATTENDANCE MONITORING SYSTEM USING FACIAL
RECOGNITION FOR SAN ROQUE NATIONAL HIGH SCHOOL

A Capstone Project
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RONALD JR. M. TESNADO
GLENN H. ACIDO
MARK ANTHONY R. ANADIA
RON ZEDRICK SYMON L. MAGALLANES
DAVE JOSEPH G. RENTORIA

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Chapter 1

THE BACKGROUND OF THE PROJECT

Introduction

Attendance tracking constitutes a fundamental element of educational administration, ensuring student accountability and adherence to institutional regulations. Traditionally, schools have relied on manual recording techniques, which are progressively considered as inefficient and susceptible to human error. With the progression of educational technology, institutions are shifting toward automated solutions, employing biometrics to enhance data precision and lessen administrative workloads.

San Roque National High School encounters considerable challenges with its existing manual attendance procedure, which consumes considerable time from teachers and administrators and results in inaccuracies monitoring school entry, exit, and classroom presence. To resolve these particular issues, this capstone project introduces a Student Attendance Monitoring System that utilizes Facial Recognition (FR) technology, customized for San Roque National High School. This system will automate the recording of attendance, integrate SMS notifications for parents, and unify student and faculty information to offer a secure alternative to the current manual procedures.

By adopting this centralized system, the institution will not only reduce the risk of misplaced or falsified records but also promote a more transparent academic atmosphere. The integration of real-time violation tracking and centralized student profiling enables administrators to quickly identify patterns of

chronic absenteeism. In the end, this technological change transforms routine administrative duties into proactive student assistance, closing the communication divide between the school and parents while creating a secure, modern campus framework.

Objectives of the Project

The main objective of this study is to design and develop a Student Attendance Monitoring System using Facial Recognition Technology for San Roque National High School during the S.Y 2026 - 2027.

Specifically, the study aims to:

1. Test the software quality of the developed attendance monitoring system.
2. Assess the level of acceptability of the proposed system among its intended end-users.
3. Assess the proposed project using the Technology Acceptance Model (TAM).

Scope and Limitation of the Project

This project involves the design and implementation of a facial recognition-based student attendance monitoring system intended for students, faculty, and administrators at San Roque National High School. The system captures and processes facial biometric data along with an SMS notification framework to manage attendance records. It functions throughout the school

campus, monitoring daily entry and exit points during regular school operations and subject-specific class times. The main objective is to improve the efficiency, accuracy, and security of attendance record-keeping, thereby replacing the manual recording process.

The system operates by matching live camera feeds against a pre-registered database, but its performance is constrained by external environmental factors such as inadequate lighting, camera positioning, changes in facial appearance, and poor image quality. Additionally, the SMS notification feature depends on stable network connectivity and pre-registered contact information. The system serves purely as an administrative tool; physical security and campus surveillance remain the responsibility of human personnel.

Furthermore, the hardware deployment is explicitly restricted to designated school gate camera networks and selected classroom entry points, meaning unmonitored campus areas will not capture student movement. The scope is strictly limited to the software processing of biometric identification and attendance reporting; it does not include the physical installation of internet infrastructure. From a data privacy perspective, the facial recognition data collected will be used exclusively for attendance verification, ensuring that student records are securely maintained and not utilized for any secondary surveillance purposes beyond the scope of school administrative tracking.

Significance of the Project

The importance of this study is to modernize the traditional attendance-taking process, reduce human error, and ensure accurate attendance recording through facial recognition of individuals entering the school. This will provide a faster, more accurate, and secure way of recording and monitoring attendance. The study will provide benefits to the following:

San Roque National High School. Modernizes the traditional attendance-taking process, reducing human error and establishing a secure, technologically advanced campus environment.

School Administrators. Helps maintain organized attendance records, significantly reducing the administrative workload required to prepare and verify manual reports.

Teachers and Staff. Provides substantial time savings by eliminating manual roll calls, allowing educators to focus on instructional time and track classroom presence accurately.

Parents. Offers real-time SMS notifications regarding their child's attendance, enabling immediate follow-up on absences, tardiness, or skipped classes.

Students. Streamlines the daily attendance routine, making the entry and classroom logging process faster and more efficient.

Project Team Proponent. Provides the developers with practical experience in applying software engineering, biometric integration, and database management skills to a real-world institutional problem.

Future Project Team. Serves as a foundational reference and baseline dataset for future researchers looking to expand or improve upon facial recognition attendance systems.

Definition of terms

To facilitate the understanding of this study, different terms are defined herein.

Attendance Monitoring System. A software application that accurately records, tracks, and manages the daily attendance or absences of individuals to automate logging and eliminate manual errors.

Biometrics. Measurable physical characteristics, specifically facial features in this project, used by the system to automatically and securely verify an individual's identity without manual intervention.

Database Management. The organized collection, storage, processing, and maintenance of institutional records, including student profiles, faculty schedules, and attendance logs, ensuring data integrity and security.

Facial Recognition. A biometric technology capable of uniquely identifying an individual by analyzing and comparing facial features against a pre-registered database.

Level of Acceptability. The degree to which the intended end-users, such as teachers, administrators, and parents, find the proposed system satisfactory, user-friendly, and effective for their daily operational and monitoring needs.

Software Quality. The overall operational standard of the developed system, focusing on its reliability, functionality, and accuracy in processing facial recognition, deploying SMS alerts, and securing data.

SMS Notification. An automated text message alert dispatched by the system to a parent or guardian's registered mobile device to provide real-time updates regarding a student's attendance status.

Technology Acceptance Model (TAM). An information systems theory used in this study to evaluate how end-users perceive, accept, and utilize the attendance monitoring technology, focusing specifically on its perceived usefulness and perceived ease of use.

Violation Monitoring. An integrated administrative feature that actively logs and tracks behaviors that breach school attendance policies, such as chronic tardiness or unauthorized campus exits.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews related literature and recent studies that provide the theoretical and technical basis for the proposed system. The materials explore the evolution of attendance monitoring systems, emphasizing the shift from manual to automated tracking. It also examines facial recognition's mechanics, benefits, and privacy concerns in educational settings. The chapter synthesizes research on SMS notification systems and data protocols, justifying their integration into a secure, efficient, and compliant attendance framework.

Student Attendance Monitoring System

Educational technology advances demand a shift from manual to automated attendance systems. A Student Attendance Monitoring System ensures precise, efficient tracking, eliminating human error, saving time, and preventing unauthorized absences. By integrating technologies like facial recognition, biometrics, and real-time alerts, it turns routine administration into an initiative for accountability and campus security. According to Reynoso et al. (2025), the implementation of a student attendance monitoring system based on RFID technology and SMS notifications replaces conventional manual recording methods. This technological advancement facilitates digital attendance tracking, enabling educators to efficiently monitor and administer attendance data through desktop platforms. The automation of presence verification significantly diminishes the administrative burden on teaching staff and conserves

instructional time. Additionally, the integration of such a system strengthens essential lines of communication between educational institutions and parents concerning student attendance and location.

Similarly, the Quirino State University Research Team (2024) asserts that an attendance monitoring system constitutes a critical tool for systematically tracking absenteeism within institutional settings. In contrast to conventional biometric devices, which yield only unprocessed data outputs, the development of a centralized web-based platform enables the integration of facial recognition inputs into standardized and interpretable reports. This automated framework facilitates efficient oversight of time-in and time-out records by administrative personnel, enhancing data accuracy, reinforcing security protocols, and mitigating opportunities for fraudulent attendance logging.

Honmane et al. (2025) examine the transformative potential of an automated facial recognition system in redefining attendance tracking within educational institutions. By integrating sophisticated computer vision and machine learning methodologies, the system enables real-time, precise attendance logging, thereby mitigating the inefficiencies and inaccuracies inherent in conventional manual procedures. Requiring minimal human intervention, it streamlines operational workflows for both administrative personnel and end users, offering a robust and scalable framework suited to contemporary academic environments. Likewise, Krishna et al. (2025) emphasize that automated attendance systems effectively overcome critical shortcomings of traditional approaches, particularly deficiencies in data reliability.

Through mobile-based deployment combined with advanced deep learning architectures, including the YOLO algorithm for simultaneous multi-face detection and Deep Convolutional Neural Networks (DCNN), the system maintains high recognition accuracy across diverse and dynamically changing environmental conditions. This automation establishes a technically sound, accurate, and operationally viable infrastructure for the modernization of attendance management in intelligent classroom ecosystems.

Face Recognition Technology

Facial recognition technology automates attendance management by achieving high accuracy, particularly through its capacity to recognize students regardless of varying distances from the camera, a capability directly relevant to handling different classroom layouts and spatial arrangements (Lateef & Kamil, 2023). Modern automated systems rely on deep learning methodologies that process live camera feeds to detect faces effectively despite fluctuations in classroom lighting and minor pose variations (Khalkar et al., 2024), ensuring the system remains practical under typical, imperfect indoor public school environments. Although traditional attendance systems have been used for decades, modern institutions are adopting these technologies to maximize instructional time and streamline daily transactions.

Additionally, local studies on mobile-based attendance monitoring systems using face-tagging technology demonstrate that automating student tracking via biometrics eliminates the risk of impersonation, such as students answering for

absent classmates, and reduces the burdensome manual roll call process typically handled by teachers in public institutions (Cabrillas et al., 2021). Similarly, a local study on the QSUM-eASys notes that integrating biometrics into web-based systems supports schools' information needs by generating understandable, uniform reports for administrators (Domingo & Ladia, 2024), directly supporting the development of an administrative dashboard for school faculty and administrators.

On the other hand, broader applications of facial recognition technology extend well beyond schools, raising significant security and privacy concerns. Hogenhout and Wangmo (2025) explored how the widespread deployment of facial recognition in public surveillance, retail, and law enforcement has led to the unconsented collection and commercialization of sensitive biometric data, underlining the need for the present study to incorporate strong security protocols to protect student records. Unlike passwords or credit cards, compromised facial data cannot be changed, exposing individuals to long-lasting threats such as identity theft and fraud (Hogenhout & Wangmo, 2025). Furthermore, the same researchers highlighted that inherent biases in training databases can cause discrimination against specific demographics, emphasizing the critical need for strict security, ethical governance, and data protection compliance when implementing facial recognition systems for minors (Hogenhout & Wangmo, 2025).

Automated Attendance Monitoring

Automated Attendance Monitoring refers to the use of technology to record, track, and manage the attendance of people with minimal manual effort. It has been adopted as an alternative to traditional attendance-taking practices, where teachers or designated staff manually record students' presence.

According to Ali et al. (2022), attendance systems are essential for recording and tracking the presence of users in various environments and ensuring that attendance is properly managed. They also discussed various automatic identification technologies developed to improve attendance management and address the shortcomings of existing approaches.

Using automated systems for attendance tracking can help streamline and manage the process. Traditional handwritten attendance records can be time-consuming, particularly when many students are involved. They can also require additional effort in maintaining and reviewing attendance records. Roy et al. (2022) pointed out that various technologies, such as biometrics, RFID, Bluetooth, GPS, and QR codes, have been used in attendance systems to overcome the challenges of traditional attendance-taking. These technologies provide different approaches to automatically recognize and record people's presence.

In educational institutions, automated attendance monitoring can also be combined with facial recognition technology to identify students and record their attendance. This makes it possible to record attendance without requiring students to sign an attendance sheet or requiring teachers to call each student's

name. Ahmad et al. (2024) developed an automated attendance system that utilizes artificial intelligence and facial recognition, where the recognized face is used to confirm a student's attendance. The study evaluated several pretrained face recognition models and determined that FaceNet was the most suitable model for the developed system, achieving an accuracy of more than 95%.

Additionally, automated attendance systems can help make attendance information more accessible and easier to manage. Electronic attendance records can be stored and retrieved for future reference and monitoring by authorized personnel. Makigod et al. (2024) developed an IoT-based automated attendance system using facial recognition technology to improve the traditional attendance-taking process. Their system was designed to provide an effective and touchless method of recording attendance. These developments show that automated attendance monitoring can contribute to the efficiency, organization, and reliability of attendance monitoring in educational institutions.

In general, automated attendance monitoring provides an effective approach to making attendance monitoring more efficient and less dependent on manual processes. The incorporation of technologies such as facial recognition enables automated systems to identify students, record their presence, and maintain attendance data while reducing manual intervention. This can help develop a more efficient attendance monitoring system that can be implemented in schools.

SMS Notification System

An SMS Notification System is essential to establish real-time communication between the school's biometric database and students' guardians. Automatically informing parents of a student's arrival, absence, or unscheduled departure ensures timely awareness, turning routine attendance monitoring into a proactive safety protocol. According to Guevarra and Corpuz (2023), deploying an attendance monitoring system is essential, as conventional methods often fail to notify parents promptly about their children's absences. In response, contemporary systems have been developed to automatically transmit real-time alerts to guardians through Short Message Service (SMS), including timestamps that record a student's arrival at academic sessions or school-related events.

Similarly, Reddy et al. (2025) developed an intelligent attendance monitoring framework incorporating real-time SMS-based notification capabilities to address the limitations inherent in conventional manual attendance procedures. The system enables immediate recording of attendance records and facilitates automated dissemination of updates to relevant stakeholders, including parents and institutional administrators, thereby mitigating instances of fraudulent attendance reporting. Empirical findings demonstrate that the implementation of instantaneous alerts substantially increases stakeholder involvement and accelerates operational efficiency.

Additionally, Bhanu et al. (2026) examine the integration of real-time SMS alert systems with sophisticated facial recognition technology to overcome

persistent inefficiencies in conventional attendance procedures, particularly those related to time inefficiency and human error. Inaccuracies in attendance recording frequently result in diminished transparency and delayed dissemination of information to parents. The proposed system automates the process by capturing student images via a webcam and instantaneously verifying them against a pre-enrolled database, thereby enabling contactless attendance registration. Upon confirmation, an automated SMS is dispatched to parents, providing immediate notification of the student's attendance status. This synthesis of intelligent biometric identification and real-time messaging infrastructure facilitates prompt communication, strengthens institutional accountability, and significantly alleviates the administrative burden on academic personnel.

Student Information and Data management

Student information and data management is the process of coordinated collection, storage, processing, and maintenance of information concerning students. In educational institutions, the management of student information is crucial since schools manage a variety of data types, such as student profiles, attendance records, contact information, and biometric data. With a well-designed data management system, authorized staff can easily access and manage student data and ensure its accuracy, organization, and security. This pertains especially to the proposed Student Attendance Monitoring System. This

system would handle student profiles, attendance records, and facial recognition data.

As stated in Republic Act No. 10173, or the Data Privacy Act of 2012, Section 2,

“It is the policy of the State to protect the fundamental human right of privacy, of communication while ensuring free flow of information to promote innovation and growth. The State recognizes the vital role of information and communications technology in nation-building and its inherent obligation to ensure that personal information in information and communications systems in the government and in the private sector are secured and protected.”

Similarly, Section 13 of the same Act provides that the processing of sensitive personal data is prohibited unless certain conditions are met, one of which is the consent of the data subject for the specific purpose prior to processing. This is applicable to the proposed system, as processing information is important for facial recognition, and the information should be well protected. The law thus establishes a valuable legal foundation to guarantee that students' personal data and biometric data are collected, stored, and processed in an appropriate manner.

Another empirical research on student information management is the big data technology-based university student information management system proposed by Feng, Zhan, and Zhao (2021). The study was targeted towards designing an appropriate architecture to minimize data duplication and achieve quick response to queries when the amount of student data grows. Similarly,

Alpasan, Maligang, and Esimos (2021) created a web-based attendance monitoring system for a public high school in the Philippines that has an SMS notification feature that automatically notifies parents or guardians whenever a student is marked absent. These studies illustrate the potential of centralized and organized information systems to enhance the management of student information and attendance as opposed to the traditional manual approach.

In addition, Coronel and Morris (2023) included other terms that were important during database design, such as normalization and data integrity, which help organize related data, minimize unnecessary data duplication, and ensure consistency and reliability of stored data. The concepts, in conjunction with the provisions of Republic Act No. 10173, serve as a basis for building a dependable Student Information and Data Management component in the proposed Student Attendance Monitoring System with the aid of Facial Recognition Technology in San Roque National High School. Student records and attendance data can be managed correctly, authorized staff can obtain relevant information efficiently, and SMS notifications can help with timely communication about student attendance through an organized database, while student profiles, attendance records, and facial recognition data can be handled concurrently.

Chapter 3

METHODOLOGY

This chapter details the methodology, software engineering paradigm, architectural designs, data flows, hardware and software specifications, ethical compliance, and evaluation mechanisms implemented in the development and validation of the **Student Attendance Monitoring System Using Facial Recognition for San Roque National High School**.

Project Development Framework

The development of the Student Attendance Monitoring System Using Facial Recognition for San Roque National High School applies the Classical Waterfall Paradigm (Linear-sequential life cycle model) as its principal software engineering methodology. As stated by Ali Khan (2023), this approach requires a strictly sequential process in which each distinct development phase from requirement analysis to system design, implementation, testing, and deployment must be fully completed before the next phase can commence. This guarantees that no overlap occurs among the various stages of system development. Because of this mandatory linear progression, the framework necessitates thorough specification and validation of all system requirements prior to proceeding with the actual design and coding of the application. In addition, Kaopiz (2026) points out that because the Waterfall model makes returning to a completed stage highly challenging, it requires rigorous initial planning and comprehensive documentation from the start. By thoroughly validating the

architecture in advance, this strict adherence to preliminary planning ensures a highly organized and predictable route for the successful implementation of the facial recognition and attendance monitoring features.

Although iterative methodologies such as Agile are common in purely cloud-based web applications, the Waterfall Model was adopted for this capstone project because of its hardware dependent architecture, strict legal constraints, and academic requirements. The system requires tight integration between physical hardware and its software pipelines.

Furthermore, the processing of biometric facial data referring to minor students requires strict compliance with the Data Privacy Act of 2012 (Republic Act No. 10173). To ensure compliance, data protection measures, parental consent procedures, and encryption hashing protocols required thorough design and validation before any software development. Finally, this phased methodology is directly consistent with the structured milestone oversight enforced by the University of Rizal System - Cainta Campus. The academic regulations of the college require distinct stage-gate reviews, formal milestone approvals, and detailed preliminary documentation, making the Waterfall model as the most suitable approach for implementation.

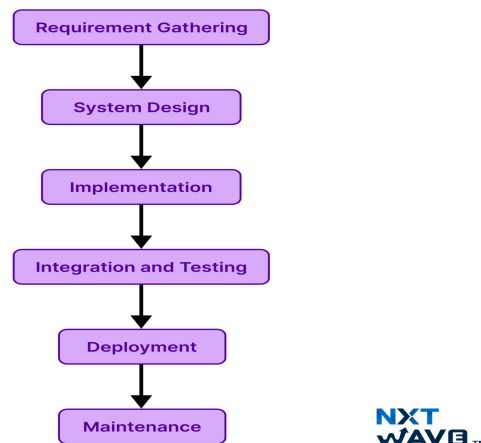
Software Engineering Paradigm

The system's development is organized according to the sequential stages of the **Waterfall Model**. This framework creates a structured development

process in which each stage provides validated outputs that act as requirements for the following phases as shown on Figure 1.

Figure 1.

Classic Waterfall model



Requirement Analysis and Gathering

The proponents conducted observation, focus group discussions, and workflow assessments involving the school administration, teaching staff, and also interviewed some of the ICT personnel of San Roque National High School. The limitations of conventional manual roll calls using pen and paper, including lost instructional time, liability to attendance tampering, and delayed guardian notifications, were recorded. These operational challenges were converted into functional specifications (real-time facial biometric identification, immediate automated SMS dispatch, and live attendance logging) and non-functional specifications (system dependability, data protection, and an inference response time under 1.5 seconds).

System and Architecture Design

Based to the specified requirements, technical schematics and modeling diagrams were developed. This stage involved creating the development of the Single-Page Application (SPA) component architecture with React, the design of the relational database structure and Row-Level Security (RLS) policies within Supabase, and the specification of the computer vision workflow for the webcam.

Implementation and System Coding

The software components were developed during this phase:

Frontend Development: An administrative dashboard was built using React bundled with Vite, utilizing React Router for page routing, Tailwind CSS for responsive styling, Framer Motion for UI transitions, and React Hook Form for student profiling and input validation.

Backend as a Service (BaaS): Implemented via Supabase, hosting a cloud-managed PostgreSQL relational database, user authentication, and real-time WebSocket event listeners.

Integration and System testing

The webcam vision engine, Supabase cloud backend, React administrative portal, and cloud SMS gateway were integrated into a fully working end-to-end system. Thorough unit testing, module integration testing, and stress testing under diverse ambient lighting conditions and angles were performed.

Deployment and Maintenance

The fully assembled terminal station was installed at the specified entry and testing area of San Roque National High School to conduct field trials. Stakeholders evaluated system performance through the Technology Acceptance Model (TAM), and modifications were made to camera exposure, log cooldowns, and biometric matching thresholds as applied.

Cost-Benefit Analysis

A Cost-Benefit Analysis (CBA) serves as a structured economic framework employed in technology projects to establish that the system's long-term value justifies its initial development costs. The five-year projection period was used because it aligns with the typical lifecycle schedule of ICT infrastructure in educational settings, including the computer workstations and webcam devices installed in public secondary schools.

Figure 2

Cost Benefits Analysis

COST

Cost Category	Year 0 (Development)	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Initial Investment							
Software & License	10,000.00	--	--	--	--	--	
Face Recognition Setup	5,000.00						
Hardware (Cameras)	10,000.00	--	--	--	--	--	
Development Team	60,000.00	--	--	--	--	--	
Testing	10,000.00	--	--	--	--	--	
System's Setup	5,000.00	--	--	--	--	--	
Workstations (Desktop &	200,000.00	--	--	--	--	--	
Recurring Costs							
Maintenance & Support		12,000.00	12,000.00	12,000.00	12,000.00	10,000.00	
Training		3,000.00	3,000.00	3,000.00	3,000.00	3,000.00	
SMS Notification Service		10,000.00	10,000.00	10,000.00	10,000.00	10,000.00	
Data Security/System Backup		10,000.00	10,000.00	10,000.00	10,000.00	10,000.00	
Total Cost	300,000.00	35,000.00	35,000.00	35,000.00	35,000.00	33,000.00	
Present Value of Cost	300,000.00	31,818.18	28,925.62	26,296.02	23,905.47	21,732.25	432,677.54

BENEFITS

Benefit Category	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Staff Time Savings (Salary 15,000)	180,000.00	180,000.00	180,000.00	180,000.00	180,000.00	
Paper & Printing Savings	10,000.00	10,000.00	10,000.00	10,000.00	10,000.00	
Manual Processing Savings	24,000.00	24,000.00	24,000.00	24,000.00	24,000.00	
Report Generation	12,000.00	12,000.00	12,000.00	12,000.00	12,000.00	
Report Correction Savings	6,000.00	6,000.00	6,000.00	6,000.00	6,000.00	
Total	232,000.00	232,000.00	232,000.00	232,000.00	232,000.00	
Present Value of Benefit	210,909.09	191,735.54	174,305.03	158,459.12	144,053.75	879,462.53

Benefit Cost Ratio=		879,462.53	2.03
		432,677.54	
Metric	Result	Decision Rule (for Project Acceptance)	Conclusion
Benefit Cost Ratio (BCR)	2.03	BCR > 1	Accepted

Summary of CBA:

Cost/Benefit Category	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Cost	300,000.00	35,000.00	35,000.00	35,000.00	35,000.00	35,000.00	475,000.00
Benefit		232,000.00	232,000.00	232,000.00	232,000.00	232,000.00	1,160,000.00
Return on Investment		28.8%	28.8%	28.8%	28.8%	28.8%	
Total Cost	475,000.00						
Total Benefit	1,160,000.00						
Total ROI	144%						
Annual ROI	28.8%						
Number of Years	5 Years						

Architecture Design

Dataflow Diagram

System Flowchart

Site Map / Site Directory

Database Schema

Technical Specifications

System Testing / Evaluation

Locale of the Project

Subject of the Project

Ethical Considerations

Procedures of the Project

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